

Innovation From Differentiation: Pollution Control Departments and Innovation in the Printed Circuit Industry

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Invited Paper

Abstract—Theory predicts that managers will create specialized boundary-spanning departments to insulate a firm from changing surrounding conditions. Theory also predicts such insulating structures will inhibit adaptation. I find that in response to changing water-pollution regulation, top managers indeed created specialized pollution-control departments to insulate the existing process, but some of these departments acted not as insulators but as conduits for information and thereby helped both to improve the efficiency of the production process and to reduce pollution. The special role of these departments allowed unique access to and perspective on information from inside and outside the organization and gave them an incentive to use this information to improve core processes. Thus my research suggests that pollution-control departments may belong to a class of boundary spanners that represent a special case for organizational design.

I. INTRODUCTION

MANY companies now face the dual challenge of tougher global competition and increasing obligation to protect the natural environment. In response, some firms, such as Union Carbide, have created "environmental" departments to mediate between the firm and environmental regulators, while others, such as 3M and Dow, have attempted to integrate responsibility for the natural environment into existing organizational roles [1]–[3]. Theory suggests that choosing between the two alternatives creates a dilemma for managers. A separate boundary-spanning department, such as Union Carbide's, allows scale efficiencies, controlled risk, and efficient production, while an integrated approach, such as 3M's, allows the firm to learn and adapt to new conditions [4]–[7]. Organizational theories predict that most managers will respond to this dilemma by forming specialized pollution control departments to protect existing procedures, processes, and structures [4].

This paper investigates the formation and behavior of pollution control departments in the printed-circuit industry. I find that top managers created separate pollution control departments to insulate the firm from regulators, but that some

of these departments did not act to insulate the firm, rather they provided important process-improving information and innovative ideas to process engineers. I find that pollution control departments were able to make these process improvements because they had unique information and that this information derived in part from their specialized role. I propose that this unexpected behavior suggests extensions of theories of organizational design.

II. THEORETICAL BACKGROUND

Pollution control departments span the boundary between the organization and its surroundings. Organizational theory provides three main perspectives on the effect and behavior of such boundary-spanning entities. Time-honored models of organizational design suggest that specialized boundary-spanning entities allow organizations to remain efficient in the face of changing business conditions, but may inhibit adaptation to changes in external conditions [4], [5]. In contrast, proponents of "lean production" or "total quality" argue that such departments reduce information flow and thus impede adaptation and improvement of process efficiency [7]–[9]. A third perspective considers the potential of boundary-spanning departments to cause organizational conflict, and thus emphasizes the need for conflict-reducing integrating mechanisms [10].

The customary view of the role of specialized boundary-spanning departments derives from theories of division of labor and efficiency [11]–[13]. These theories assume that focused application of skill and technology best leads to efficiency and learning [11], [13]. In this view, lateral interactions between tasks or departments require application of scarce management resources and distract attention from optimizing the task at hand [4], [14]. If changing conditions (such as increasing environmental regulation) require the firm to create new tasks to address new conditions, these tasks should be performed in separate organizational structures so that they do not disrupt the production process (which is assumed to be optimized). Thus rational firms seek "to isolate their technical cores from environmental influence by establishing boundary-spanning units to buffer or level environmental¹ fluctuations"

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¹Thompson did not, of course, mean to indicate only the natural environment but also the general conditions surrounding the firm.

[4, p. 67]. In this perspective, since environmental influence can only reduce efficiency, rational organizations attempt to minimize information processing costs between boundary-spanning departments and the core [4], [14]–[17].

In the customary view, such separation and specialization of boundary-spanning groups maintains efficiency but hinders adaptation. Burns and Stalker [5] argue that reduced division of labor, reduced specialization, increased lateral communication, and increased task integration allow more rapid recognition of new conditions, modification of objectives and acquisition of new capabilities. Nevertheless, Burns and Stalker [5] share the perspective that in stable environments lateral communication patterns and task integration increase communication and coordination costs without supporting performance improvement.

Another perspective arises from proponents of “quality” and “lean production” who suggest that separation and specialization of boundary-spanning groups hinder *both* efficiency and adaptation. Unlike Thompson [4], Galbraith [14], and even Burns and Stalker [5], who assume that efficiency arises from the focused application of specialized skills and technology, proponents of “quality” or “lean production” assume that efficiency results from the lateral exchange of information and broadening of incentives [7], [8]. This efficiency can be accomplished, they claim, only by distributing responsibility for tasks (such as quality and pollution control) throughout the company. Only when this responsibility is shared, do workers have the incentive and the ability to transfer relevant information, and only when such information transfer occurs can organizations make processes more efficient or adapt to new conditions [8, p. 99].²

The above perspectives consider information transfer to be central to the influence of structure on performance. In contrast, Lawrence and Lorsch [10] emphasize the role of structure in fostering or impeding intergroup conflict. Lawrence and Lorsch [10] argue that precisely because boundary-spanning departments mediate between the organization and different surrounding institutions, these departments develop differing goals, time perspectives, interpersonal orientations and formality of structures. These differences then cause conflict between groups which must be offset through the use of integrative devices (such as meetings and teams) [10, p. 12].

In the remainder of this paper, I outline several hypotheses consistent with the above theories, and attempt to validate them by exploring the behavior of boundary-spanning departments that managers of firms in one industry created in response to environmental regulation. I show that the above theories do not accurately predict the extent, nature, or implications of interaction between these boundary-spanning departments and the organizational core. I then propose extensions to theory to explain this unusual behavior.

²The “quality” movement has had the greatest influence on professionals in industry and analysts in government policy. Bemoski [18], Thomas [19], and Martin [20] argue that pollution control and quality departments have analogous roles in the organization. Therefore, they expect pollution control groups to buffer the firm from a changing environment, delay and filter feedback information, and prevent core-process adaptation [3], [21]. The Congressional Office of Technology Assessment, in its seminal report, *Serious Reduction of Hazardous Waste*, argues that pollution control departments will insulate the firm from environmental requirements, and consequently prevent core-process change that might reduce the production of hazardous waste [22].

A. Hypotheses

Thompson [4] predicts that, when faced with changing external conditions, rational organizations form separated boundary-spanning departments to mediate between the firm and its surroundings. While “quality” theorists and industry analysts propose that separate pollution control departments will impede information transfer and learning, they agree with Thompson [4] that firms will respond to changing environmental conditions by creating such departments, and they expect the relationship between these departments and the rest of the organization to be distant [7], [8].³ Piore and Sable [6] also anticipate that US firms will respond to changing surrounding conditions by creating such boundary-spanning departments. Once these departments are created, Thompson [4], Galbraith [14], MacDuffie [7], and Womack *et al.* [8] predict that relations between these departments and the rest of the organization will be distant. Thompson [4] and Galbraith [14] predict that interaction with such departments will relate predominantly to issues of task coordination. Thus my first two hypotheses are:

Hypothesis 1: In response to changing environmental conditions, managers will create pollution control departments to insulate the firm and maintain the efficiency of core processes.

Hypothesis 2a: Pollution-control departments and core technical and managerial personnel will have little interaction.

Hypothesis 2b: Interaction between pollution-control and core personnel will relate to environmental issues.

Thompson [4] and Galbraith [14] both predict the core organization will perform best when boundary-spanning departments effectively insulate the core from surrounding conditions. In this model, boundary-spanning departments allow the core to focus on process efficiency. Thus:

Hypothesis 3a: Communication between pollution-control departments and the process engineering staff will lead to process change that improves environmental performance.

Hypothesis 3b: Interaction between pollution-control departments and the process engineering staff will result in process change that increases cost, reduces quality, or limits the capability of the production process.

Hypothesis 3c: In high performing firms (low cost, high quality) pollution-control departments will communicate less frequently with process-engineering personnel than in low performing firms.⁴

III. RESEARCH METHOD

Understanding issues of structure, communication, and performance requires multiple research methods to compare multiple organizations. To allow practical research, one must choose between rich, multimethod data and a large sample size. I chose to identify a few similar manufacturing organizations

³In part for this reason, government has begun to create legislation that “does not stop at the boundary of the firm,” and instead requires the adoption of pollution control accounting and control practices throughout the firm [23].

⁴It might also be possible to form an alternative hypothesis from the quality literature that interaction with the environmental department will be associated with process innovation that improves environmental performance and improves core performance. This literature generally assumes, however, that such communication is unlikely [8].

and then analyze them in detail using qualitative interviews, archival records, and several written surveys. I believe that in combination, these multiple sources triangle organizational behavior.

A. The Research Setting

The printed circuit fabrication industry is a rich domain in which to study response to environmental regulation and the behavior of pollution control departments. The industry includes many companies that use similar technology to produce commodity products, thereby allowing cross firm comparison of performance. In the early 1980's, printed circuit firms faced more stringent regulation of the discharge of metals and organic compounds. Most of these regulations became standardized around the time of the promulgation of federal standards (40CFR413-June 15, 1983).

All of the plants in my study produce multilayer rigid printed circuit boards. All of the companies make medium complexity boards for the computer industry, and fewer than 10% of their production volume (measured in panels) is of high complexity.⁵ Sales vary from \$5 million to 50 million. Two plants are owned by large computer firms; the rest represent the chief activity of a closely held firm.

B. Qualitative Research Method

1) *Interviews*: To understand how printed circuit fabricators have responded to new water pollution requirements, I investigated the history and behavior of eight plants in detail. I began with open-ended interviews of firm presidents, plant managers, process engineers, quality personnel, and pollution control personnel. To focus the field of inquiry, I concentrated on the etching and plating part of the production process—usually known as the “wet-process.” Engineers responsible for this part of production are called Wet Process Engineers.

After several exploratory interviews, I inquired about the history of the plant and organization, and walked through the plant with staff from the departments of wet process engineering (WPE), quality, and pollution control (PC). During each of these tours, I asked that major process changes be identified. I also developed a list of potential improvements and sought to determine if firms had discovered and adopted these improvements.

2) *Archival Analysis*: To uncover responses to water pollution regulation, I investigated government, institutional and company archives from the 1970's and 1980's. Sewer authorities inspect each company at least eight times a year and give extensive descriptions of technology and practice.

C. Quantitative Research Method

The quantitative research measures the extent, nature and result of interaction between PC departments and core process personnel. Twelve rigid printed circuit firms agreed to participate in the written study and were sent questionnaires for

⁵ Defined as 1) density of three traces between 0.100" centers, or 2) surface mount with fine pitch applications.

the wet process, quality, and pollution control managers. Eight firms returned all three surveys, two returned two surveys, and two did not respond. I also sent performance surveys to the 12 firms with questions about quality, production cost, and treatment cost. I received responses from seven.

1) *Measuring Interaction*: To measure the nature of interaction between functional groups, the survey asked managers of the wet process, quality, and pollution control departments to characterize both their own and their compatriots' behavior. The survey asked questions about time spent communicating with personnel from other departments in the firm, the nature of interaction and the topics discussed.

2) *Measuring Process Innovation*: The managers of wet process engineering, quality, and pollution control were asked to report the five most important process changes that had occurred in the last year and to rate the members of each of the three groups in terms of their role in: 1) “identifying the opportunity or need that became the impetus for the change,” and 2) “in designing the change.” The managers were then asked to indicate the effects of these process changes. Respondents could choose one or more of the following options: improved product quality, improved production capabilities, reduced production costs, improved pollution control, or other/don't know.

3) *Firm Performance*: The performance survey separated boards of unusual complexity from commonly produced ones. For both types of boards, it measured yield rates at several stages and the cost of materials used in production and in treatment. Because of differences in cost tracking methods at the firms, I was unable to use the responses to questions about the cost of production materials.

IV. INITIAL RESPONSE TO WATER POLLUTION REGULATION

Early in my investigation, I found support for the hypothesis that firms will create specialized departments to respond to changing conditions (Hypothesis 1). In eight firms where I performed intensive interviews and archival analysis, managers responded to increased water pollution regulation by purchasing waste filtration equipment and forming a pollution control department to protect the firm from regulatory problems.

In response to new environmental regulations, a senior member of the technical staff and a top manager of each facility selected the technology and organizational structure to use. Top managers remembered the decision as necessary to reduce the risk of regulatory trouble. All of the managers chose to buy robust technology and sized this equipment to include substantial overcapacity. All eight companies sized treatment equipment to between 1.5 and three times *that needed for growth*.⁶ Consider for example, the decision of the president of one company:

“[The plant] presently discharges approximately 20 000 gal/day...or 40 gal/min. Average production has been determined to be 643 sq/ft/day. Allowing for at least 50% increase in production and, consequently, water usage over

⁶ One plant sized the facility at “30% over peak load,” and one company that purchased a new building inherited a treatment system with three times the capacity they expected to need.

the next three to four years, the treatment system would have to be sized to handle a maximum flow rate of 60 gal/min. However the [president] has required that the system be designed to handle a maximum flow of 100 gal/min."

Managers seem to have purchased large, robust treatment systems in a conscious attempt to reduce uncertainty by precluding regulatory problems. As the president of one company put it, "Make my industry stable, and I'll look to making my industry better." All but one company purchased a ferrous sulfate clarifier to remove metal contaminants from waste water. Although more efficient and effective technologies were available, ferrous sulfate clarification provides reliable and fluctuation-insensitive waste treatment.⁷ All of the top managers initially charged the pollution-control personnel with treating and controlling waste at the boundary of the plant. As one manager put it, "It's easier to pay someone to operate a waste-treatment plant than put up with meddling in manufacturing." Pollution control managers at all of the plants agreed that in the early 1980's their job was (as one put it) to "treat what production sent them."

Thus as suggested by Hypothesis 1, managers chose to create pollution control departments to insulate the firm from changing external conditions and to maintain the efficiency of core processes.

A. Evidence of Interaction Between Pollution Control and the Rest of the Firm

In numerous interviews, I found behavior that contradicted the hypothesis that communication between these specialized departments and the organizational core would be infrequent and limited [Hypothesis 2(a)]. While in about half the firms PC personnel reported seldom conversing with wet process personnel, in the other firms they judged such conversations to be frequent. Three classes of behavior for pollution control departments emerged (Fig. 1).

My qualitative evidence also contradicted the hypothesis that interaction between pollution control departments and the organizational core would lead to poorer performance [Hypothesis 3(b)]. In discussions with wet process and pollution control managers, I learned that pollution control departments often had information important to improving the production process in areas unrelated to environmental protection. Thus when WPE managers acquired this information, substantial performance improvement could result. In one firm, PC personnel had initiated innovations that, over one year, caused a 41% reduction in the use of process materials (an annual saving of \$113 000 or about 1% of sales).

1) *Interaction Between Pollution Control and the Rest of the Firm:* Contrary to Hypothesis 2(a) (pollution control departments will seldom interact with the organizational core), my data indicate that PC personnel interact extensively with the rest of the organization. WPE managers reported speaking more often (though not significantly so) to PC personnel than to maintenance, management, purchasing, or other engineering

⁷One manager jokingly calls this the "American" method, since it is nearly foolproof, requires little skill to operate, and waste-treatment operators can independently solve most problems simply by adding more ferrous sulfate.

Classification And Description Of PC Departments In The Eight Firms Participating In Case Studies And Intensive Interviews	Number of firms in class
Information Conduits: The pollution-control department attempts to reduce waste as well as treat it. PC personnel frequently pass information on performance, problem areas and solution ideas to process engineers. Quote: "They (PC personnel) are like a second pair of eyes and ears for me (the process manager)"	2
Feedback Sources: The pollution-control department insures that the firm is in compliance by treating waste, but it should not have to treat waste unnecessarily. If a problem occurs in production, PC personnel pass relevant data to production concerning water flow, chemical composition and so forth. Quote: "If something goes wrong, we (Pollution-Control) let them know about it."	2
Buffers: The pollution-control department insulates the production process from regulatory change by treating waste. PC personnel inform the organization of environmental regulations and get involved in production issues only when treatment or health issues arise. Quote: "We (Pollution-Control) should stay out of process procedures."	4
Total of 8 departments classified	

Fig. 1. Three classes of behavior for pollution-control departments.

personnel (usually dry process engineering). Moreover, despite the current importance of quality in the industry, PC and quality managers did not differ significantly in total time spent communicating with the rest of the organization ($N = 9; \mu_{pc} = 12.7$ hrs/week $= \sigma = 8.65$ $\mu_{qual} = 17.7$ hrs/week $\sigma = 14.2$). Quality managers did differ significantly in time spent communicating with "production personnel" and "management" ($p < 0.05$). This difference may be caused by management concern about quality and by the direct role quality managers often play in certain functions in production (e.g., automatic optical inspection).

The data also indicate that conversations with PC personnel frequently include topics other than pollution control [in contrast to Hypothesis 2(b)]. While both pollution control and wet process managers reported conversations about "waste treatment" as "very common," conversations about "waste of material in production" were reported by both as "common." PC managers reported that their own conversations with WPE personnel commonly included discussion of "quality" and "yield."

2) *Pollution Control Interaction and Process Innovation:* If interaction is measured by the effect of the pollution-control department on core technology, the data again show that PC personnel interact with the organizational core [in contrast to Hypothesis 2(a)]. WPE managers rated their own personnel as most important in initiating and designing process changes (see Table I), but judged quality and PC personnel to be of nearly identical importance in initiating process changes.⁸

Remarkably, in two firms the Wet Process Engineering manager rated PC personnel as more important, on average, than his own staff in initiating reported process changes.

3) *Pollution Control Interaction, Process Innovation and Firm Performance:* Hypothesis 3(a) predicts that pollution control interaction with the core production process will be associated with process innovation that leads to improved environmental protection. The data support this hypothesis.

⁸Data about process innovation from the three managers exhibit a substantial respondent bias. All three types of managers tended to report innovations in which their own personnel played a key role. Since WPE managers can best judge innovations in their department and likely will underestimate the role of other departments, I concluded that WPE manager reports of innovation would be informed but conservative.

TABLE I
WET PROCESS ENGINEERING MANAGERS RATINGS OF IMPORTANCE
OF PERSONNEL OF DIFFERENT GROUPS IN INITIATING
MOST IMPORTANT PROCESS CHANGES IN PREVIOUS YEAR

Department	Mean Rating of Importance	Standard Deviation of Rating	N Firms	N Process Innovations
PC	0.99	0.72	8	33
Quality	1.193	0.59	8	33
WPE	1.45	0.51	8	33

Anova for blocks: $F_{\text{block}} = 1.77$; $F_{\text{group}} = 0.76$; $Df_g = 2$; $Df_b = 7$;
 $Df_e = 14$; $p_b = 0.8$; $p_g = 0.47$

Scale: 0 = not at all important, 1 = important, 2 = very important

As reported by the wet process manager, communication time with pollution control personnel was associated with PC initiated process innovations (Kendall's Tau $t = .62$, $p < .1$), and 84% of these changes resulted in pollution reduction.

In contrast to Hypothesis 3(b), the data also show that PC initiated innovation often led to improvement of the production process relative to core objectives. PC initiated innovations were as likely to lead to cost reduction as to pollution control and often resulted in quality improvement. Out of 33 reported innovations, PC personnel received the highest importance rating for 13. Of these, 84% resulted in pollution reduction, 92% in cost reduction, 61% in quality improvement, and 38% in extension of production capabilities. Obviously, innovations often had numerous outcomes. Surprisingly, WPE managers reported that not a single innovation that originated from pollution control resulted in only pollution prevention improvement; all had some other benefit.

Data from five firms (from which all surveys were received) provide some suggestion that, in contrast to Hypothesis 3(c), interaction between pollution control and process engineering departments may improve the performance of the core process. Three firms that reported PC initiated quality innovation had 8% higher first pass yields for etched layers (an important measure for quality) than two firms reporting no PC initiated quality improvements.⁹ Likewise, WPE managers reported discussing quality (along with other nonpollution control topics) with PC personnel more frequently in three firms with high quality than they did in two firms with low quality (as measured by first pass yields for etched layers).

B. Causes of Interaction Between Pollution Control and the Rest of the Firm

Thus as predicted, firms did indeed form boundary-spanning (pollution control) departments to buffer themselves from changing regulatory requirements. However, contrary to expectations, these departments had extensive interaction with the core of the firm, and this interaction was correlated with performance improvement with regard to both environmental protection and core performance.

One might argue that such behavior is the result of underlying firm characteristics that really caused the changes in innovation and performance. For instance, pollution control communication rates could result from an "organic" organ-

ization, where lateral communication is emphasized. Alternatively, pollution control personnel might have unusual individual characteristics (e.g., better education or more recent exposure to another company). The data do not show evidence, however, that the WPE-PC communication patterns result from individual or firm level characteristics. Organizational levels of communication, as estimated by the wet process and quality managers, are not correlated with the volume of communication between the WPE manager and PC personnel (Fig. 2). Nor is the total time wet process and PC managers spend communicating with the rest of the organization correlated with how often they communicate with each other (Fig. 2). Additionally, PC managers had somewhat shorter tenure in the firm (3.9–6.8 years) but longer job tenure (3.5 years as opposed to 1.9) [Neither difference is significant, even at $p < .1$]. Three PC managers reported obtaining chemistry or chemical engineering degrees, whereas six WPE managers reported having chemical engineering degrees.

My research suggests that pollution control managers, rather than inhibiting process improvement, actually can provide useful information and support. My quantitative data show that it was not communication *per se* but rather communication about certain topics that led to innovation. Communication about topics outside of the normal purview of pollution control best explained PC initiation of process innovation. Two breadth measures comprised of 1) quality and yield related topics and 2) scrap and efficiency topics both correlate strongly with PC initiated innovations (Fig. 2). Likewise, breadth correlates with communication time, and communication time correlates with PC innovation. The partial correlation between breadth and innovation, controlling for communication time is still high ($t = 0.70$). The partial correlation between communication and innovation, however, is much lower ($t = 0.34$).

The clear correlation between the breadth of conversation topic and communication time could suggest either that frequent communication leads to a broadening of discussion topics, or that knowledge about a variety of topics encourages more frequent communication. To further explore this relationship, I constructed a scale of PC "information helpfulness" from reports of 1) PC frequency in requiring (for pollution control reasons) process changes and 2) PC frequency in providing process improvement information.¹⁰ I hypothesized that pollution control departments that were more likely to provide useful information than new demands would communicate more frequently with WPE personnel. As predicted, this scale is correlated with PC-WPE communication volume ($t = .65$, $p < .05$).

My quantitative data closely match observations from my qualitative study. I often observed PC personnel passing non-pollution control information to process engineers. Likewise, process engineers frequently asked pollution control personnel for information on production chemicals or problems. Several interviews with PC managers were interrupted by process engineers or personnel who were seeking information or assistance.

⁹In an industry as competitive as this such a difference can easily mean survival.

¹⁰The scale was constructed by subtracting scores for frequency of requiring change from scores for frequency of providing helpful information.

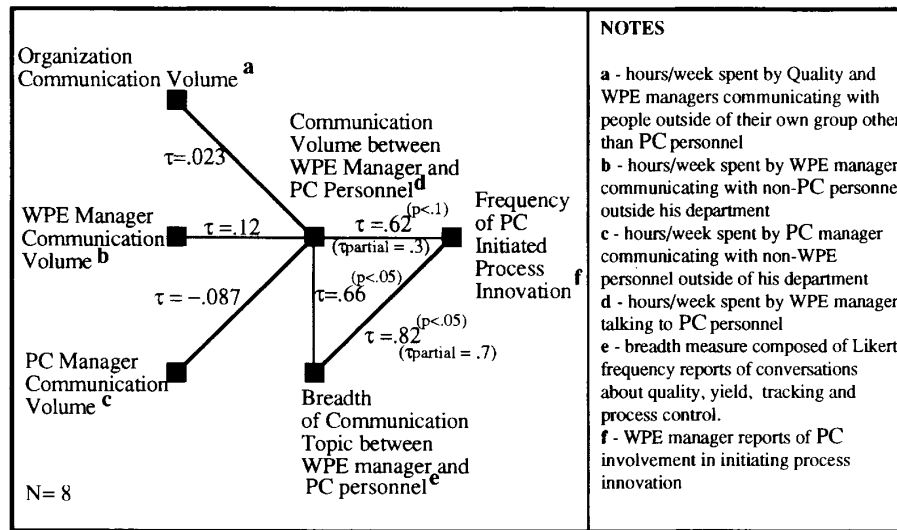


Fig. 2. Paired Kendall Tau correlations between communication measures and PC initiated process innovation.

TABLE II
COMPARISON OF PC AND WPE REPORTS OF DISCUSSION
TOPICS WITH PERSONNEL FROM THE OTHER GROUP

Topic	Mean Difference Between Reports	Std. Dev.	DF	Prob
Quality	-0.72	2.08	8	0.33
Yield	-0.39	1.50	8	0.46
Waste of Material in Production	0.17	1.77	8	0.78
Scrap	0.056	1.18	8	0.89
Waste Treatment	0.89	1.05	8	0.035

Thus I believe that pollution control departments actually have unique and important information that can be used to improve the core of the firm. When process engineers tap this resource, core process improvements result. Such a claim leads to an obvious question: Why do pollution control departments have information that is useful in improving the production process? My research uncovered several explanations.

C. Why Pollution Control Has Unique Information

1) *Different Perspective on Commonly Available Information:* Pollution control personnel work in a forgotten part of the production process. They are concerned with discarded material, byproducts of the operation, mistakes and waste. While process engineers often focus on one process quality measure (scrap rate laminar layer), pollution-control managers must treat waste from every process and thus must consider a wider range of waste sources. Perhaps because of this, pollution control personnel bring a different perspective to production problems. They have a broader definition of yield that includes yield on chemicals, copper anodes, and cleaning materials. Their definition of "quality" is also more general and includes the reliability of process baths, the predictability of variance and the safety and welfare of workers.

The quantitative data also support these assertions. PC and WPE managers disagreed about the frequency with which

they discussed certain topics with each other (Table II). The data suggest that different perspectives on quality led WPE managers to report that more conversations related to "waste treatment" and pollution-control personnel to report that more related to "quality" or "yield."

By bringing a different perspective to process problems, and by looking at different process signals, pollution-control personnel may be able to extract unique information about process problems.

2) *Internalizing Externalities:* Pollution-control departments must process material that has been discarded. In doing so, they extract information contained in waste. In all of the firms that I visited, PC managers knew that this waste could be used to diagnose problems in the upstream process. Most commonly PC personnel told me that if the water in the treatment process turned blue, it meant that chelate copper was leaking from one of the pumps in production. Since most companies process their cleaning water continuously, PC personnel must mindfully respond to process variation and thus may be unusually aware of production fluctuations.

3) *Waste Provides Diagnostic Information:* By reverse engineering, pollution control departments can extract the location of problems from waste. Just as a urinalysis provides rich information about human health, production waste provides extensive information about the health of the production process. More than many process signals, the waste stream carries information about the location and cause of process inefficiencies. For instance, at one firm pollution control engineers noticed that they frequently received stacks of discarded boards that had been partially etched and plated but not electrically tested. Upon investigation they found that the plating operation performed best when the hoist contained a full load. To ensure the best performance, the ends of production jobs were padded with extra boards to the nearest full load. Extra boards were then discarded. Wet-process and PC personnel designed a system of reusable dummy boards.

As a result, the firm avoided discarding 5000 boards per year at an estimated savings of \$25 000 (0.25% of sales).¹¹

4) *System-Wide Information:* Pollution-control personnel have unusually diverse sources of information that allow them to gain system-wide perspectives on problems. They gather data in the form of waste from stages of production that cross regions of authority and technology. For instance, at one local firm the PC personnel noticed that several different groups (the chemical lab, production, and lithography) would discard out-of-date chemicals of the same type at approximately the same time. Moreover, they noticed that the supply of these chemicals in the stock room fluctuated widely. They investigated and found that simultaneous orders from separate production centers gave purchasing an incorrect impression of increased demand, which in turn caused an overly large order and an increase in inventory. To solve the problem, the PC engineers developed a computer tracking system and a purchasing projection system.

5) *Access to Outside Information:* Pollution control engineers have unique opportunities to access information from outside of the company. In some cases, regulation causes them to gather and store particularly detailed technical information. The rapid pace of technological change also causes PC personnel to interact frequently with outside vendors. In addition, PC personnel have unique access to information from chemical vendors. As one person put it: "When production calls up a vendor, they get connected to sales. When we call up, we get connected to the lab. They know that we have to know what is in it [the vendor's product] in order to treat it."

Pollution control personnel also engage in "know how" trading of information with competitors and suppliers of chemicals [24], [25]. Vendors arrange tours of interesting waste treatment processes, and many firms welcome visits of waste treatment processes from other companies. In Massachusetts, local pollution control managers are now forming an informal association that will meet monthly to discuss technical issues.

6) *Motivation to Improve the Production Process:* Managers of pollution control departments have an incentive to use any useful information that they acquire to improve the core process. For instance, one PC manager argues that "almost anything that I can do to help the production process helps me [to treat waste]." Another argues, that "the more I can get the process under control, the better I can reduce waste treatment." Thus some PC managers believe that they can improve their own performance by reducing waste generated in the production process and by reducing swings in production performance. As one of them said, "I don't mind the FM [variance in effluent caused by production volume changes], but I sure hate the AM [variance caused by mistakes]." Thus PC managers have created tracking and control systems for production baths, and automated purchasing and inventory systems. Such systems allow PC managers to more closely optimize their own treatment process.

¹¹This savings represents just the cost of discarded boards and does not include the cost of labor or materials needed to bring these boards to this production step.

V. IMPLICATION OF THESE RESULTS

My research suggests that pollution control departments, in the course of conducting their work, acquire information and innovative ideas that can be useful to the rest of the organization. When they communicate this information, performance improves relative to pre-existing (cost, quality, etc.) and new (environmental performance) criteria.

My research suggests an additional contingency for organizational design. It suggests that personnel performing tasks that control process waste have unusual incentives and unusual information sources, and as a result, organizational structures may influence them in unusual ways.

First, my research suggests that Thompson's concepts of interdependency and organization may be contingent on the task being performed. Tasks that control core-process waste, even if highly interdependent with core processes, may improve rather than harm the efficiency of the core. Pollution control departments create new interdependencies with and constraints on core processes (and thus should reduce performance), but these constraints may act as a "process control" to ensure minimum standards of efficiency. Moreover, pollution-control personnel often work actively to improve process efficiency. They capture information in discarded material that otherwise would have been lost and use this information to improve the efficiency of the core process.

Second, my research supports the general findings of Lawrence and Lorsch [10] that high differentiation and high integration lead to high performance in changing environments. My research suggests, however, that specialization and differentiation need not lead to conflict, as Lawrence and Lorsch [10] assume, but can instead cause a type of organic solidarity based upon expectations of mutual benefit [26]. Boundary-spanning departments that mediate core-process waste have strong incentives to reduce conflict with the core personnel so they can improve the efficiency of the core process and thereby reduce waste. Such boundary-spanning departments naturally create bridges (communication forums, etc.) to the organizational core. Thus interdepartmental integration may not require managerial support but may instead arise naturally as different departments seek to gain access to useful information and innovation sources.

Third, my research suggests that organic structures are not required to facilitate information flow, cross functional innovation, and adaptation. In fact, it poses an intriguing question. Might the initial specialization of pollution-control departments represent a useful stage in adaptation? In their separate state they can acquire new perspectives and skills that then help the organization to adapt. If so, my findings support and extend research on the development of particular process innovations. Just as Kazanjian and Drazin [27] and Tyre [28] suggest that separation and subsequent reintegration of innovation projects leads to better innovation, I suspect that immediate integration of pollution control tasks would have prevented the accumulation of specialized information, and unusual perspectives and skills and thus inhibited innovation.

Thus boundary-spanners that control process waste may pose a special case for organizational design. Because of their

interdependence with core processes, they share efficiency goals with the core organization and they acquire useful process information. Because of their separation, they have new perspectives on old problems and can innovate. For both reasons, they help improve the core's efficiency and speed its adaptation to new conditions.

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